09-30 第1篇

阅读理解（说明文） How Are Clouds Formed — and Why Do They Float in the Sky?

① Look up on any ordinary afternoon, and you will almost certainly see **clouds**. Some are soft and white, like heaps of cotton; some are long and thin, like pale brushstrokes; others, on stormy days, are dark and heavy. We speak so easily of "blue sky and white clouds" that we rarely stop and ask two simple questions: **how are clouds made, and why do they stay up there instead of falling to the ground?** The answers reach deep into the physics of our everyday world.

② The first important fact is that clouds are **made of water** — in the form of **tiny liquid droplets** or, at high altitudes, **small ice crystals**. They are **not** made of "cloud stuff", and they are certainly not gas. A cloud, seen up close, is simply a huge collection of water pieces, each so tiny that it is almost invisible to the human eye. A typical fair-weather cloud may contain **millions of tonnes of water** — but spread so thinly across a large volume of sky that it can still float gently overhead.

③ Where does this water come from? Every day, the Sun heats the surface of oceans, rivers, lakes, wet soil and plant leaves. Liquid water on the surface turns into an invisible gas called **water vapour**, which rises into the warm air near the ground. **Warm air can hold more water vapour** than cold air; in fact, this is why a hot summer afternoon feels "sticky" even before the rain. As long as the air stays warm, the water vapour remains invisible.

④ The real magic begins when this **warm, wet air rises** into the sky. As the air moves higher, it enters **colder** layers of the atmosphere. Just as warm air can hold a lot of water vapour, **cold air can hold much less**. At a certain point, the rising air becomes **too cold** to keep all its water vapour in the invisible gas form. The extra water has to go somewhere — and so it **condenses**, turning back into tiny liquid droplets. Each droplet forms around a **very small "seed"** — a grain of dust, sea salt, pollen or smoke — too small to see on its own but essential for the first droplet to build itself on. These droplets, gathered in the millions, are exactly what we see as a cloud.

⑤ This is also why clouds come in such **different shapes**. When warm air rises slowly over a wide area, the cloud spreads into flat, thin layers — these are **stratus clouds**, the kind that often bring grey, drizzly days. When warm air rises quickly and strongly in a narrow column, the cloud piles up into a big, puffy shape — these are **cumulus clouds**, the cotton-like ones of sunny summer afternoons. If a cumulus cloud keeps growing

higher and higher until it reaches a great height, it becomes a **cumulonimbus** — a tall storm cloud that can produce thunder, lightning and heavy rain.

⑥ But one puzzle remains: if a cloud contains millions of tonnes of water, **why doesn't it crash to the ground?** The answer is that each cloud droplet is **incredibly tiny** — about **0.01 mm** across — so light that the **small upward movement of air** around it is enough to keep it afloat. Think of a feather in front of a gentle fan: the fan's breeze easily balances the feather's weight. A cloud droplet is much, much lighter than a feather. Only when these droplets **join together** — many of them bumping and merging — do they grow big enough to fall as **rain**. A raindrop is about **5 000 times heavier** than a single cloud droplet; it has finally become too heavy for the rising air to hold, and it falls.

⑦ High up in the sky, where the air is extremely cold, something slightly different happens. The water droplets **freeze** into small ice crystals. These crystals make up the thin, feather-like **cirrus** clouds that look like brushstrokes near the horizon. If the ice crystals join and grow heavier, they eventually fall. In very cold weather they reach the ground as **snow**; in warmer weather, they melt on the way down and arrive as **rain**.

⑧ So the next time you lie on the grass on a bright autumn afternoon and watch a cloud drift slowly above the playground, remember: you are looking at **millions of tiny water droplets**, each built on a **speck of dust**, each light enough to be carried by the air, each a **small witness** to the long, silent journey of water from the surface of a distant sea, up into the sky, and — if we wait long enough — back down to us as rain.

注释： droplet 小水珠；water vapour 水蒸气；condense 凝结；stratus/cumulus/cumulonimbus/cirrus 层云/积云/积雨云/卷云；speck 微粒

根据文章内容，选择最佳答案。

1. According to Paragraph 2, what are clouds actually made of?

A. A special gas called "cloud stuff".
B. Tiny liquid water droplets and, at high altitudes, small ice crystals.
C. Only invisible water vapour.
D. Large pieces of solid ice.

2. What does Paragraph 3 say about warm and cold air?

A. Cold air holds more water vapour than warm air.
B. Warm air can hold more water vapour than cold air; on a hot summer afternoon this is why the air feels "sticky".
C. Warm and cold air hold exactly the same amount of water.
D. Only cold air holds water vapour at all.

3. How do the first tiny water droplets form when warm wet air rises (Paragraph 4)?

A. They are blown out of tall buildings.
B. Warm rising air cools, cannot hold all its water vapour, so the extra vapour condenses into tiny droplets that form around small "seeds" like dust, sea salt, pollen or smoke.
C. They are made by airplanes.
D. They are dropped from space.

4. Why do clouds not fall to the ground, even though they contain millions of tonnes of water (Paragraph 6)?

A. Because clouds are glued to the sky.
B. Each cloud droplet is only about 0.01 mm and very light, so the small upward movement of air is enough to keep it afloat; only when droplets join together do they become heavy enough to fall as rain.
C. Because clouds are filled with helium.
D. Because the Sun pushes the clouds up.

5. What is the main image the writer leaves us with in the last paragraph?

A. That clouds are a kind of pollution.
B. That when we lie on the grass and look at clouds, we are seeing millions of tiny droplets, each built on a speck of dust, each a small witness to the long journey of water from a distant sea up into the sky and back down to us.
C. That clouds are actually just paint in the sky.
D. That students should stop looking at the sky.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "tiny liquid droplets or, at high altitudes, small ice crystals"
【解析】云由微小水滴或高空冰晶构成。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "warm air can hold more water vapour ... hot summer afternoon feels 'sticky'"
【解析】暖空气含水蒸气更多。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "condenses ... very small 'seed' — a grain of dust, sea salt, pollen or

smoke"
【解析】凝结需以微粒为核。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段 "about 0.01 mm ... small upward movement of air ... join together ... fall as rain"
【解析】小水滴够轻，合并后才下落。A、C、D 均错。故选 B。

5.
【答案】B
【题型】主旨/观点题
【定位】第⑧段 "millions of tiny water droplets ... built on a speck of dust ... small witness ... journey of water"
【解析】末段以诗意画面收束。A、C、D 均偏离。故选 B。